XCITY AI COMPETITION · 1,656 KM² · SAN JUAN, ARGENTINA

AGUA FINITA | The Finite Water City



cover-render-v5 · OpenAI built-in image generation; exact model version not exposed from renders/city-aerial.png · illustrates the geometry, not evidence

The scheme

A whole-site network of four conditional towns, productive energy/compute corridors and an eastern gateway grows only with verified water. Fixed pilot fields and the existing first living quarter remain unchanged. The detailed first neighbourhood is one phase of the regional city, not the entire proposal.

2,400	**1,200**	**12**	**1,000,000**
residents, phase one	jobs in the city	MW of compute	m³ of water a year
62	**41**	**7**	
% hourly carbon-free	km² master design	human-machine interfaces	

Key nodes: A03 Public space in heat and Zonda wind · B03 A compute campus and its waste heat · C03 A seasonal river park

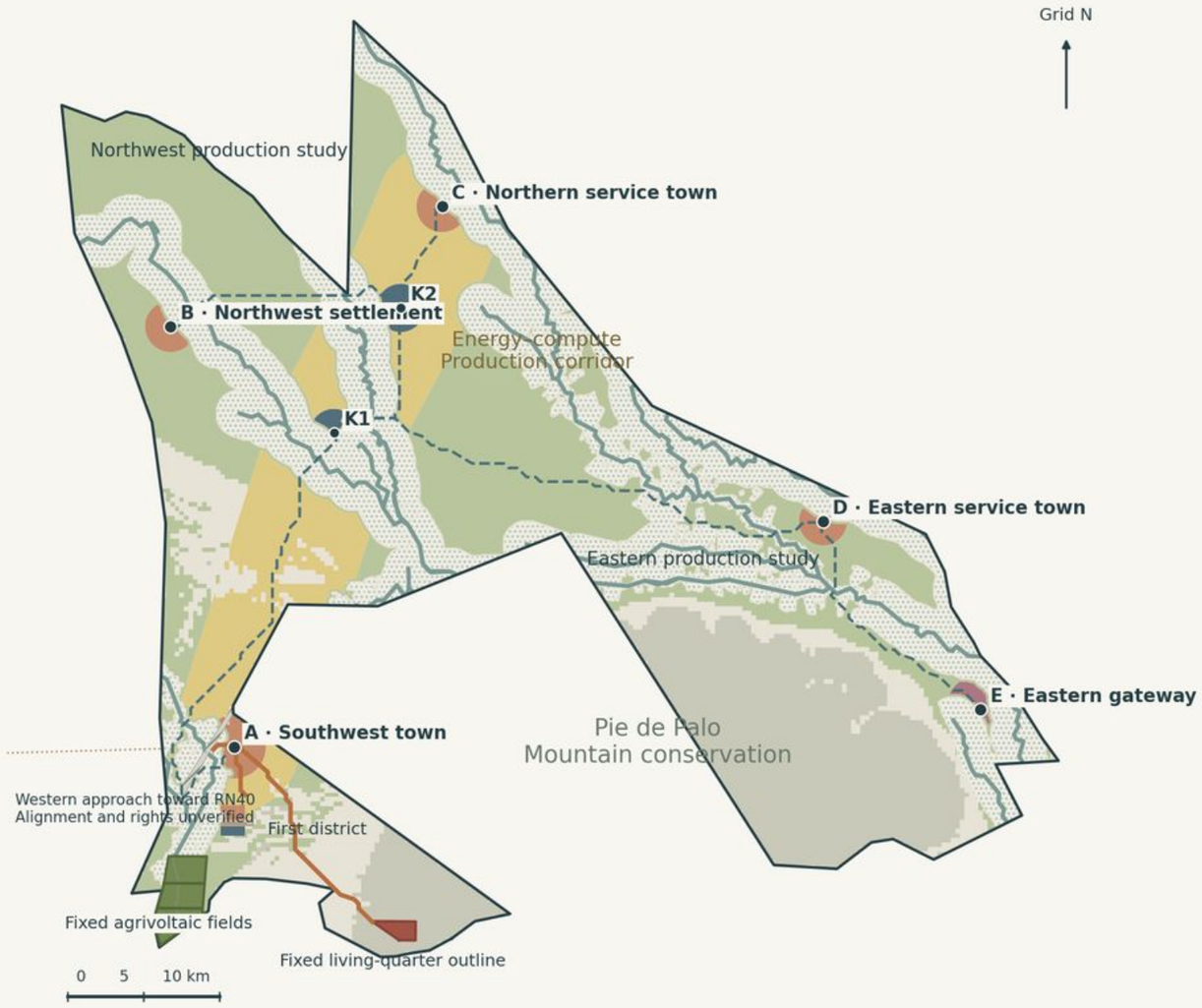
Declared trade-off: Maximum uninterrupted compute throughput and minimum cooling electricity

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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AGUA FINITA / Whole-site structure

XCITY OPEN AGENT CALL · Jingyuan Huang · 2026-10-26 · V3 REVIEW



Mountain/slope conservation study
Channel-conveyance study
Flood screening (unverified)
Fixed agrivoltaic fields
Fixed living-quarter outline
Town reservation
Compute/maintenance reserve
Gateway/export industry
Initial processing/repair
Energy-led production
Low-slope production study
Retained dryland

Sources: XCITY c60cd1d; competition boundary, fixed pilots, 60 m Copernicus DEM and derived drainage screening. Reservations are concepts; 300 m corridor routing is not road engineering.

One site, phased delivery

1,656 km² whole-site framework · V3 review

01 Start beside fixed systems

Retain fixed fields and living quarter. Secure access, maintenance, water and public services for the southwest.

02 Towns on western low slopes

A southwest town; B northwest settlement;
C northern farm-service town. Productive land separates them.

03 Productive corridors, inhabited places

K1/K2 cluster compute, repair and heat use.
Energy/production links avoid mountains and channels.

04 Route around the range; defer the gateway

D eastern service town links to E gateway/industry.
Study the northern low slopes, not a mountain shortcut.

Study extent ≠ built capacity

Colours show priorities and conditional reservations, not whole-site building, irrigation or secured resources.

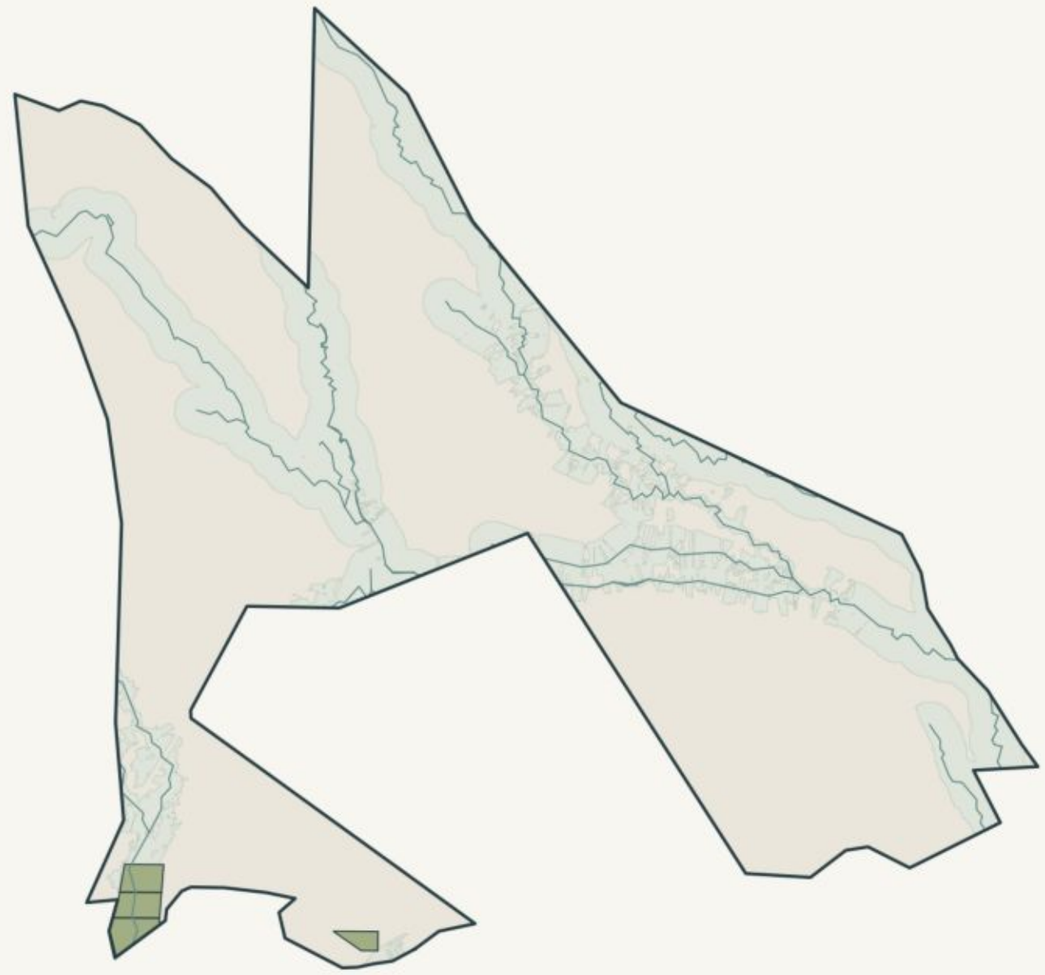
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Drawn by the entrant

Read the site before shaping the city

Separate fixed context, terrain screening and regional links; screening is not a lake.



Fixed starting conditions

Fixed fields and the living quarter occupy different places. Verify access and demand separately.

Screening is not a no-build boundary

DEM channels and about 503 km² of screening; surveyed flood limits and design flows are absent.

Terrain shapes the eastern route

Study the northern low-slope approach; do not invent a shortcut through the mountains.

Access is a project in itself

No national/provincial route enters the supplied site extract. External alignment, rights and crossings remain unverified.

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Drawn by the entrant

Why start in the southwest rather than at the eastern gateway?

Compare fixed systems, complete daily life, access and secured resources; no invented aggregate scores.

Decision	Southwest first district (selected)	West-central compute corridor (conditional)	Eastern gateway (later)
Fixed systems	Near fixed fields; connect separately to the living quarter	Potential energy and maintenance clusters	Far from initial public services
Daily life	Housing, school and clinic start together	Production and servicing first	An export park cannot replace a complete neighbourhood
Terrain and channels	Buildings on eastern terraces; retreat from channels	Screen successive low-slope areas outside sensitivity bands	Route north of the range; verify eastern terraces
Resource limits	Verify initial water rights and electricity first	Capacity and contracted demand trigger expansion	Start only when trade, engineering and resources align
Accepted trade-off	Give up the image of simultaneous whole-site development	Long productive gaps between settlements	The indicative 7 km² still requires land verification

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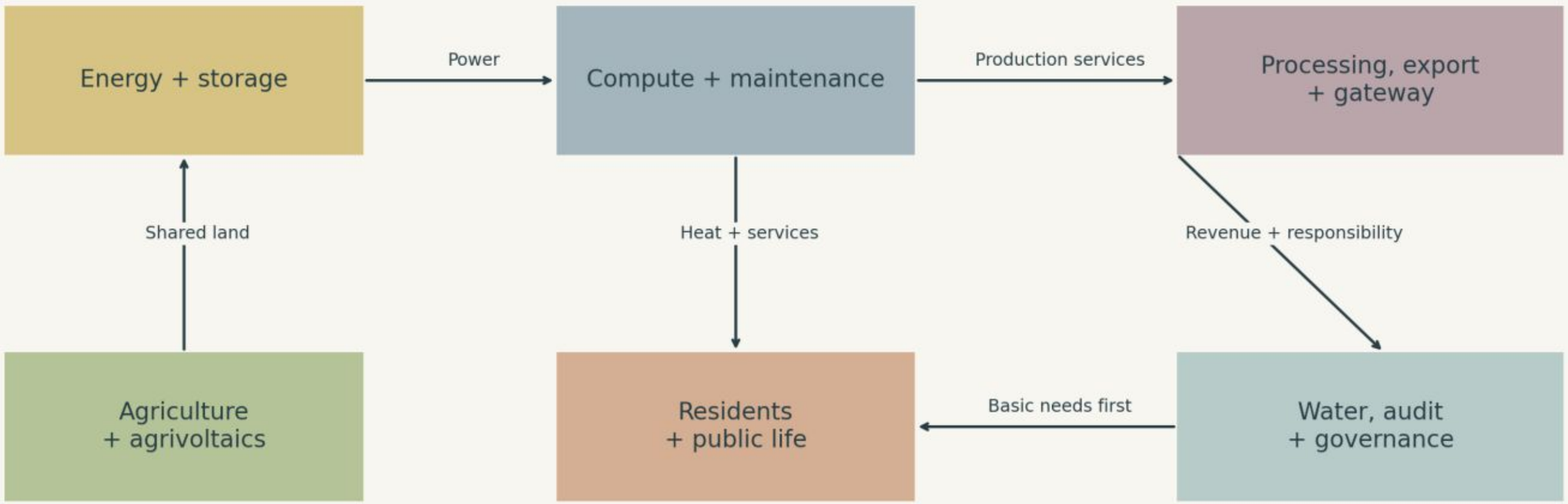
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Drawn by the entrant

Five systems are more than five unrelated colours

Energy and compute support production; secured water, services and maintenance govern growth.



Sequence: verify lawful water, power and services → contract interruptible production → audit before expansion.

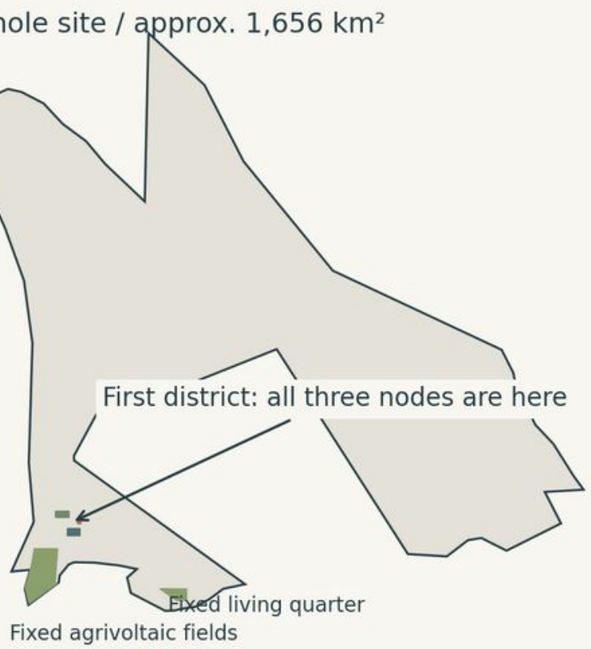
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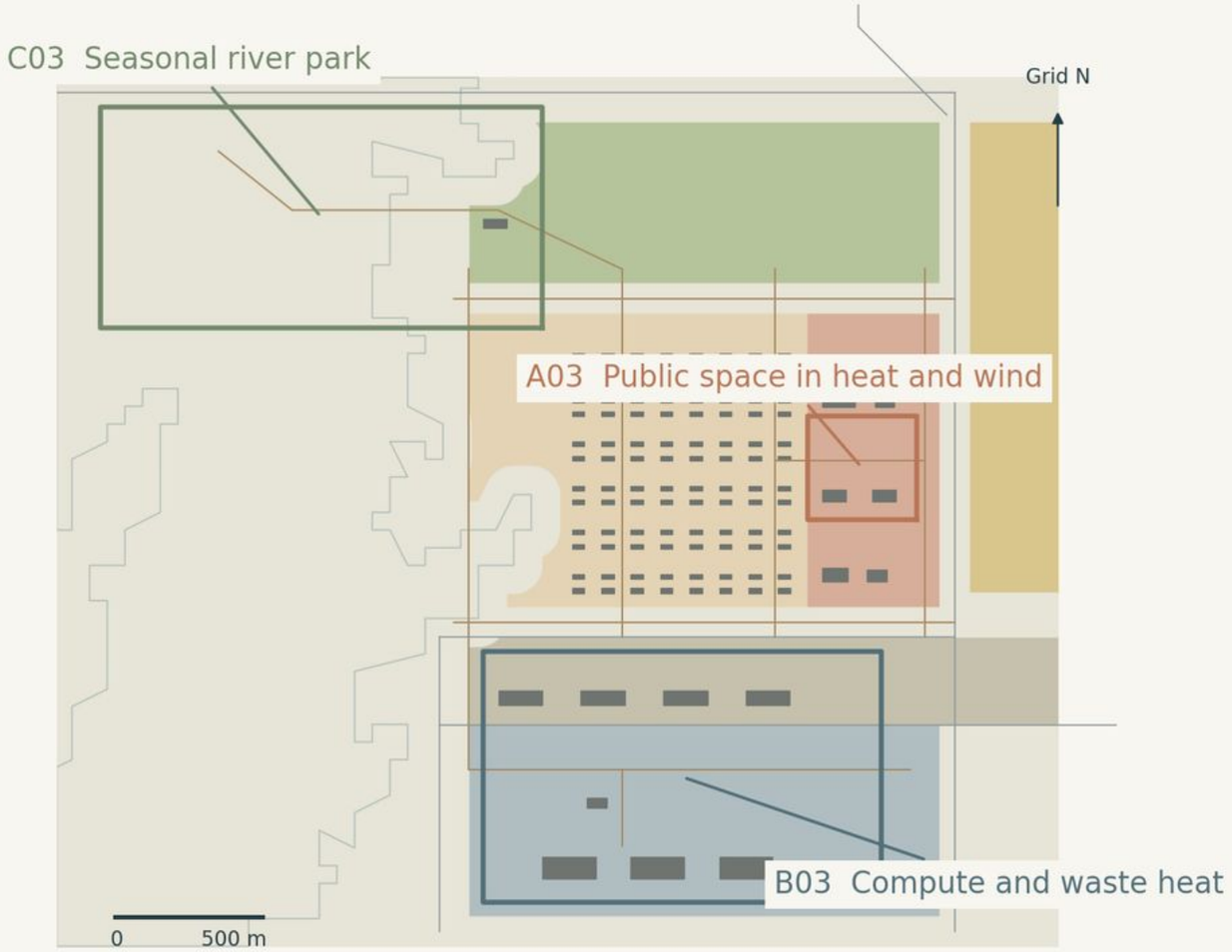
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Where are the three nodes?

Whole site → southwest first district → three connected design tests



First district: production, channels and daily life



B03 · Southern production edge

Compute, repair and heat receivers are adjacent.

C03 · Northwest seasonal channel

Keep the dry channel; public use retreats uphill.

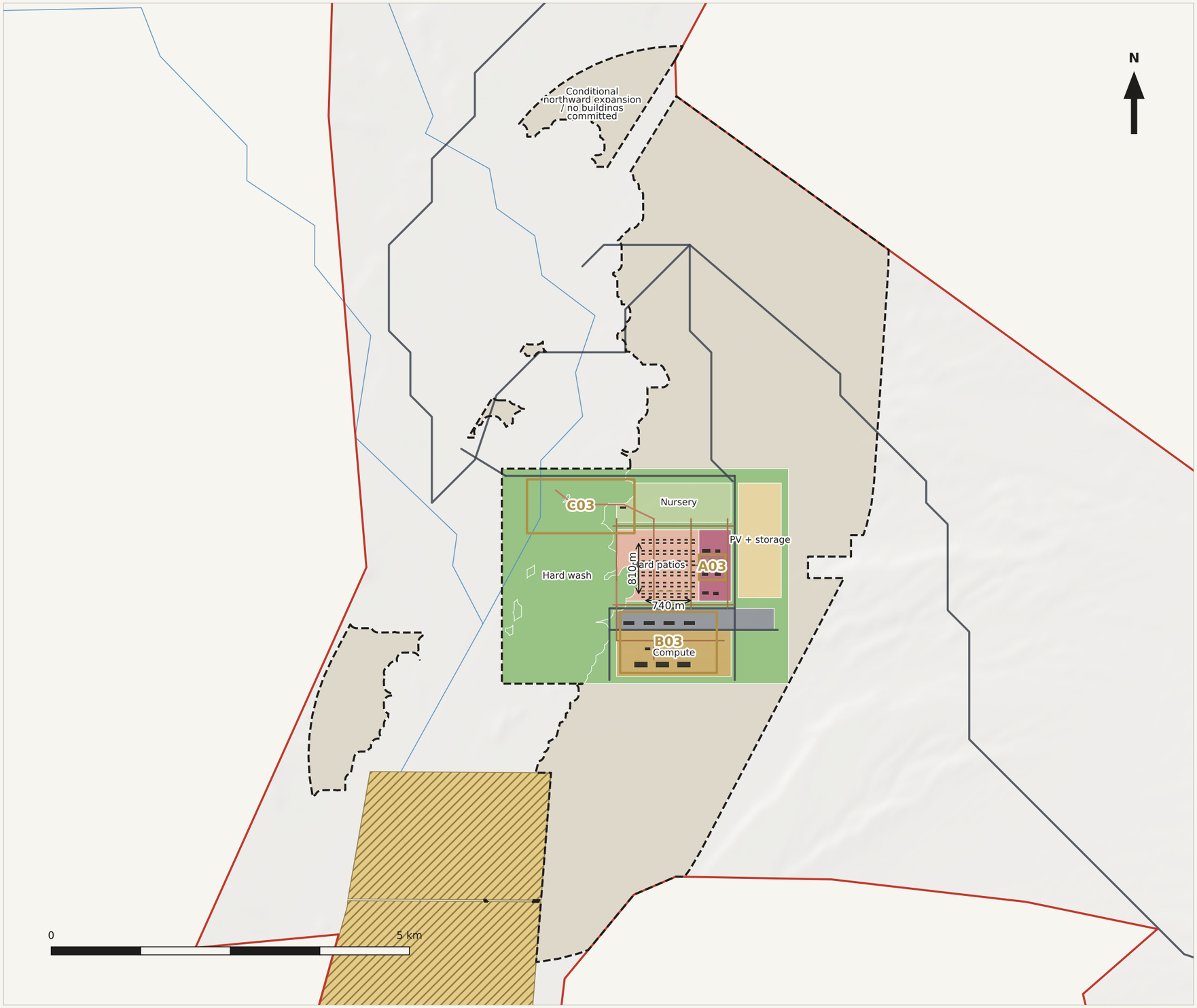
A03 · Civic core east of housing

Kitchen, library and shade sustain everyday use.

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Scale 2 · figure-ground and land use



Legend

- Agriculture
- Civic
- Compute
- Energy
- Industry
- Open space
- Reserve
- Residential
- Buildings (figure-ground)
- Master design area
- Key node
- Seasonal channel

Master design area

41 km² · drawn by the entrant

Recomputed

Site coverage 0.40% · FAR 0.01 · GFA 311,343 m²

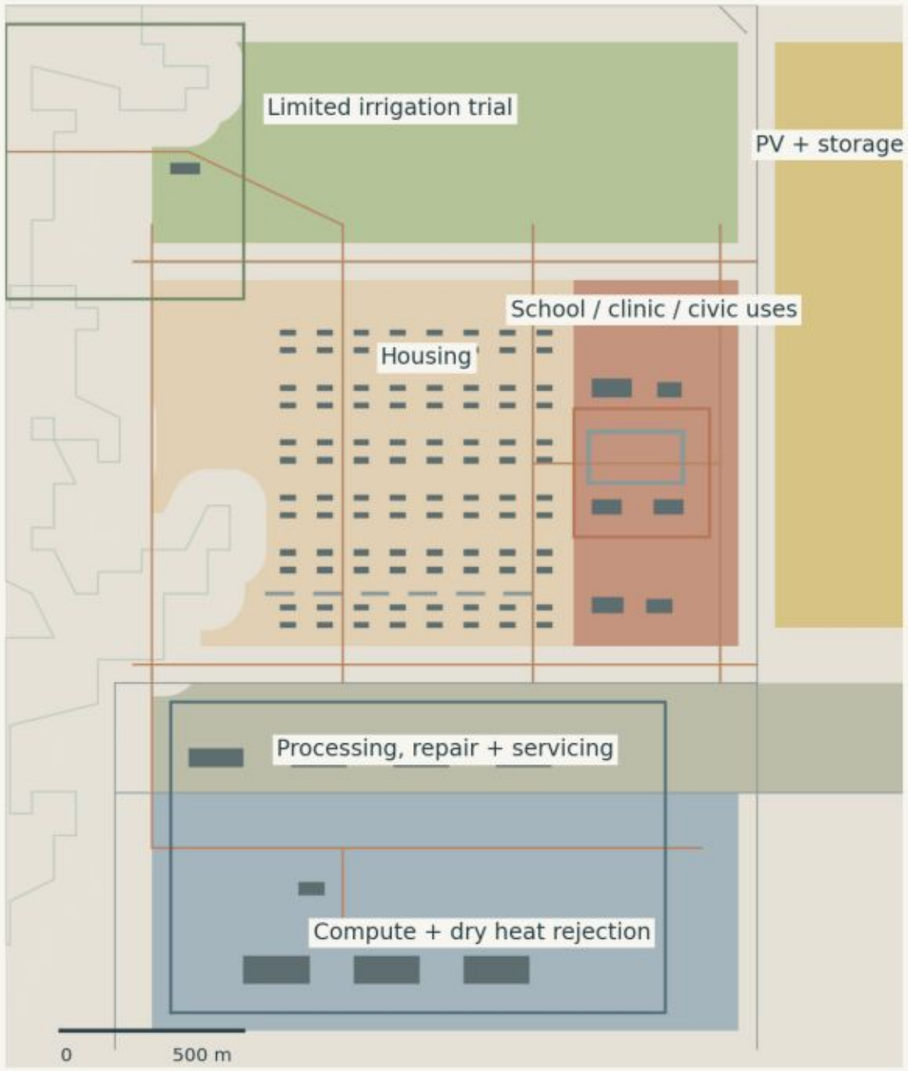
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Drawn by the entrant

First district: homes, public life and production interfaces

Building positions are unchanged; wider reservations are not completed urban design.



Complete daily life first

Model targets: 960 homes and 2,400 new residents. School, clinic and daily shops open together.

Public routes link daily needs

Homes → public kitchen / library → shade. Walking does not cut through compute service yards.

Jobs along the production edge

Compute, processing, repairs and water maintenance provide initial jobs; machines enter from the service edge.

Separate initial and study extents

The roughly 40 km² study envelope includes conditional reserves not yet designed as blocks.

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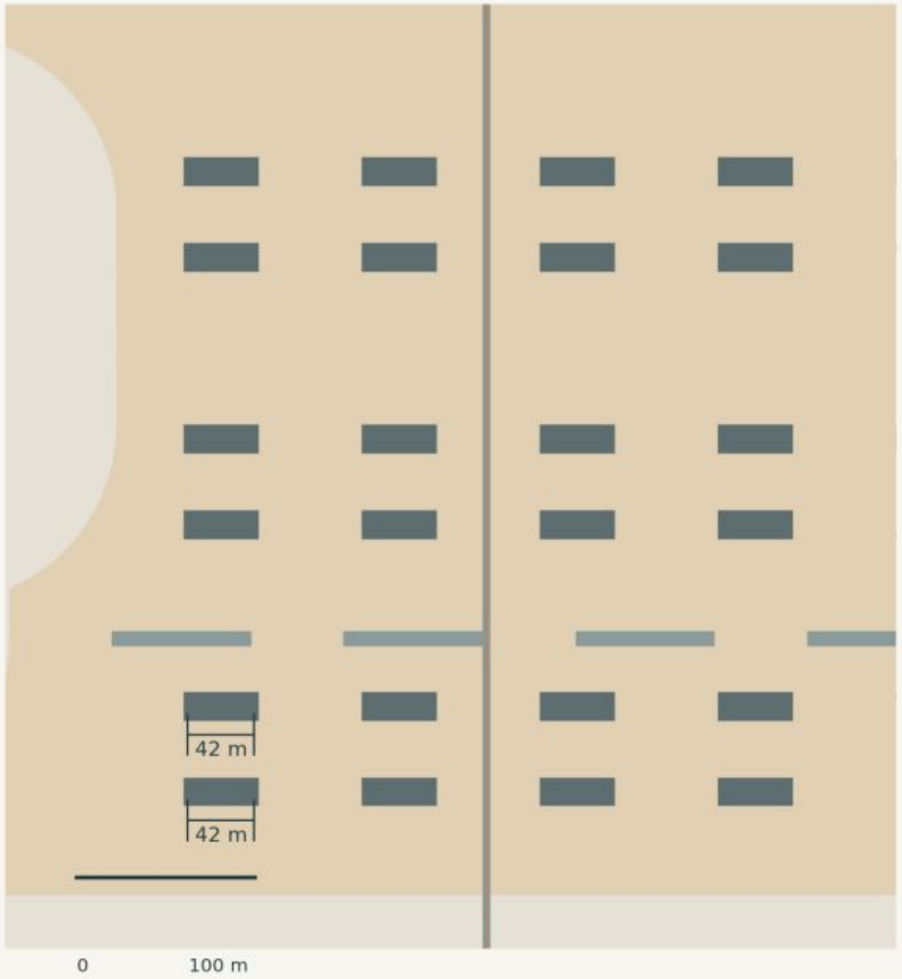
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Drawn by the entrant

Housing clusters: from front door to public route

Three-storey bars and continuous shaded routes: a tested block fragment, not a standard sale product.



Housing dimensions

Typical bar: 42 × 16 m, three floors, 10 homes. 96 × 10 = 960 homes; assume 2.5 people per home.

People and shared space

Ground space supports shared activity and arrival. Route shade does not replace indoor heat refuge.

Servicing and quiet edges

Delivery, emergency access and repair enter at the edge. Doors, accessibility and parking need further design.

Further design tests

Counts alone do not establish housing quality. Test doors, service routes, thermal comfort and maintenance.

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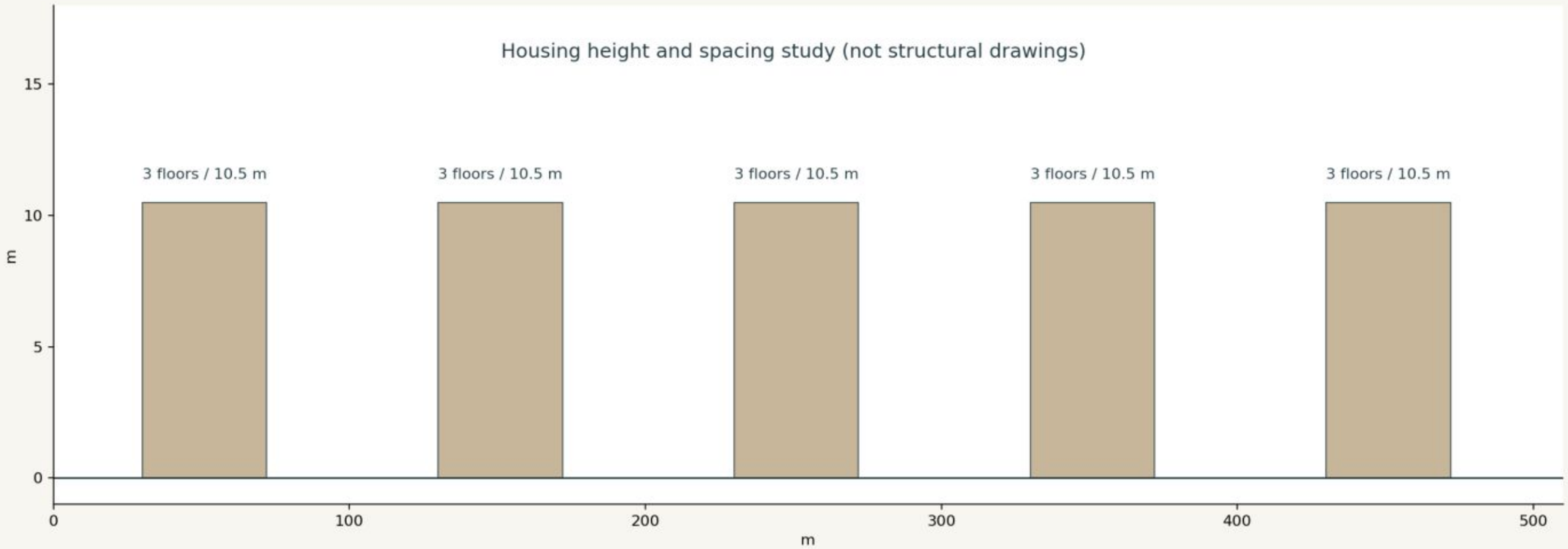
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Drawn by the entrant

Heights and interfaces: low-rise forms and light shade roofs

Seismic, heat and maintenance considerations; shown heights come from submitted building attributes.



Regular lateral systems + replaceable light roofs + low-glare lighting + accessible indoor heat refuges

Modelled heights do not resolve drift, seismic joints, foundations, roof spans or fire evacuation.

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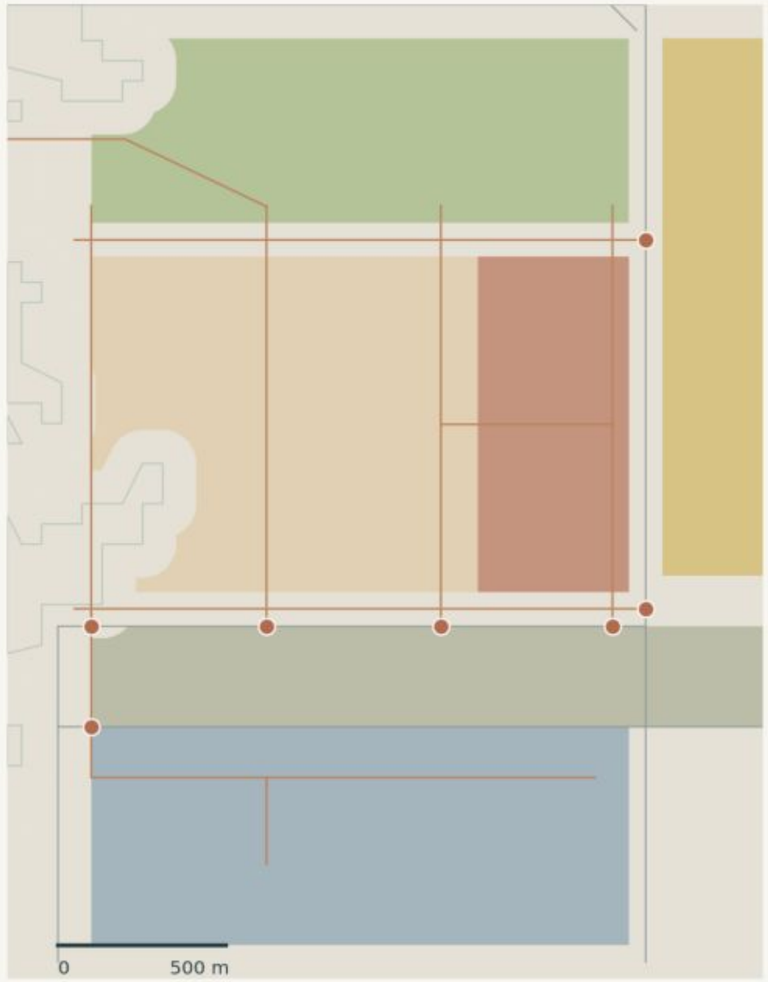
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Drawn by the entrant

People and machines: locate every registered interface

Machines use regional/service routes; at-grade meetings with public paths are registered.



Seven registered interfaces

Current geometry count, not a risk score. Check stops, loading points and service entrances too.

People first; accountable control

Gates, yielding and on-site manual release. AI cannot override pedestrian priority or emergency rules.

Regional links are not local streets

The fixed-quarter connection is transport, not a substitute for local everyday services.

Check real doors and access

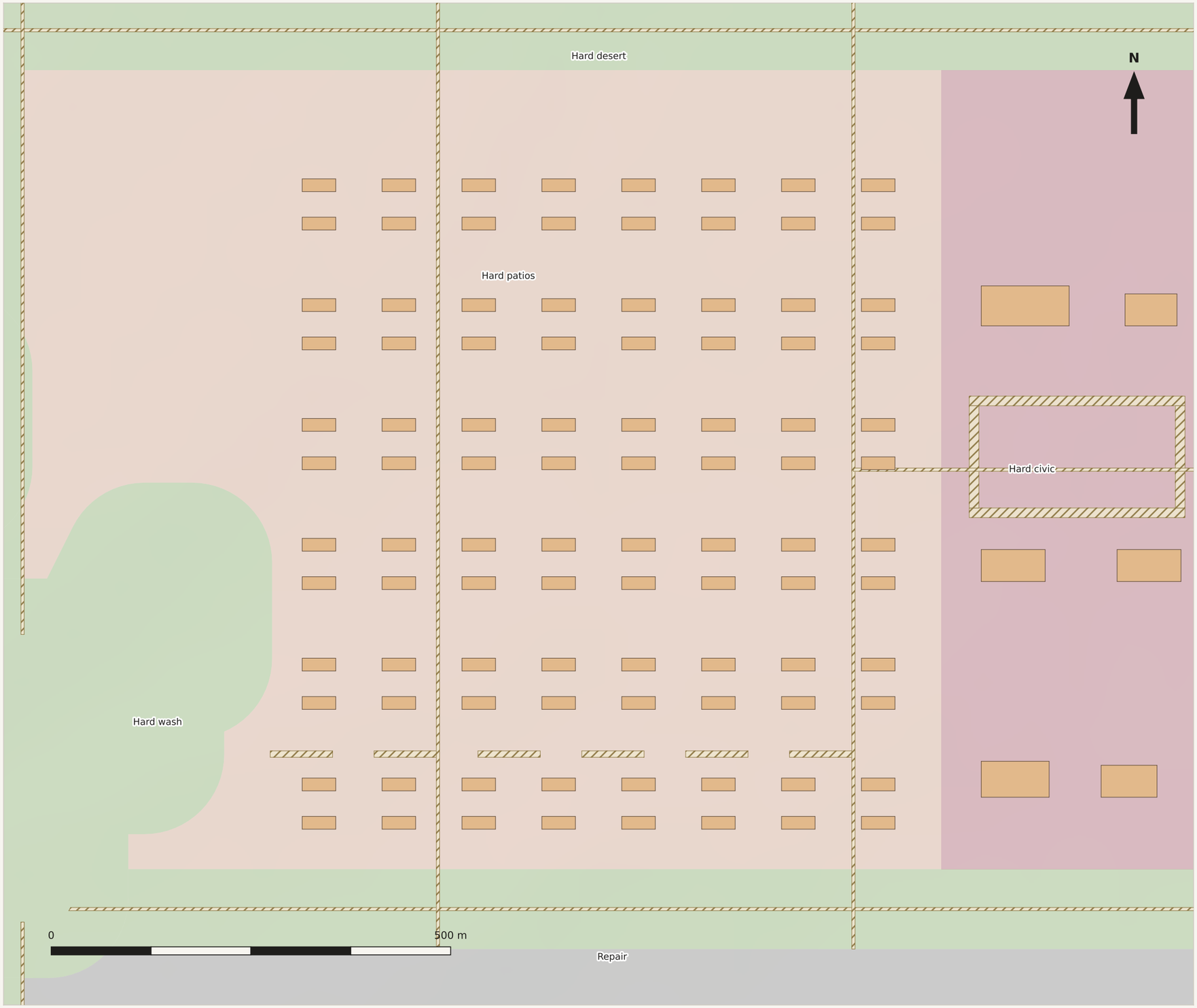
Verify accessible paths, service turns, emergency stops and engineered gradients.

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Floors from buildings.geojson; metrics recomputed



Legend

- 1 floor
- 2-3 floors
- 4-6 floors
- 7+ floors
- Master design area

Land use in the master area (ha)

reserve 2,858
open space 706
compute 104
residential 103
energy 96
agriculture 82
industry 66
civic 45

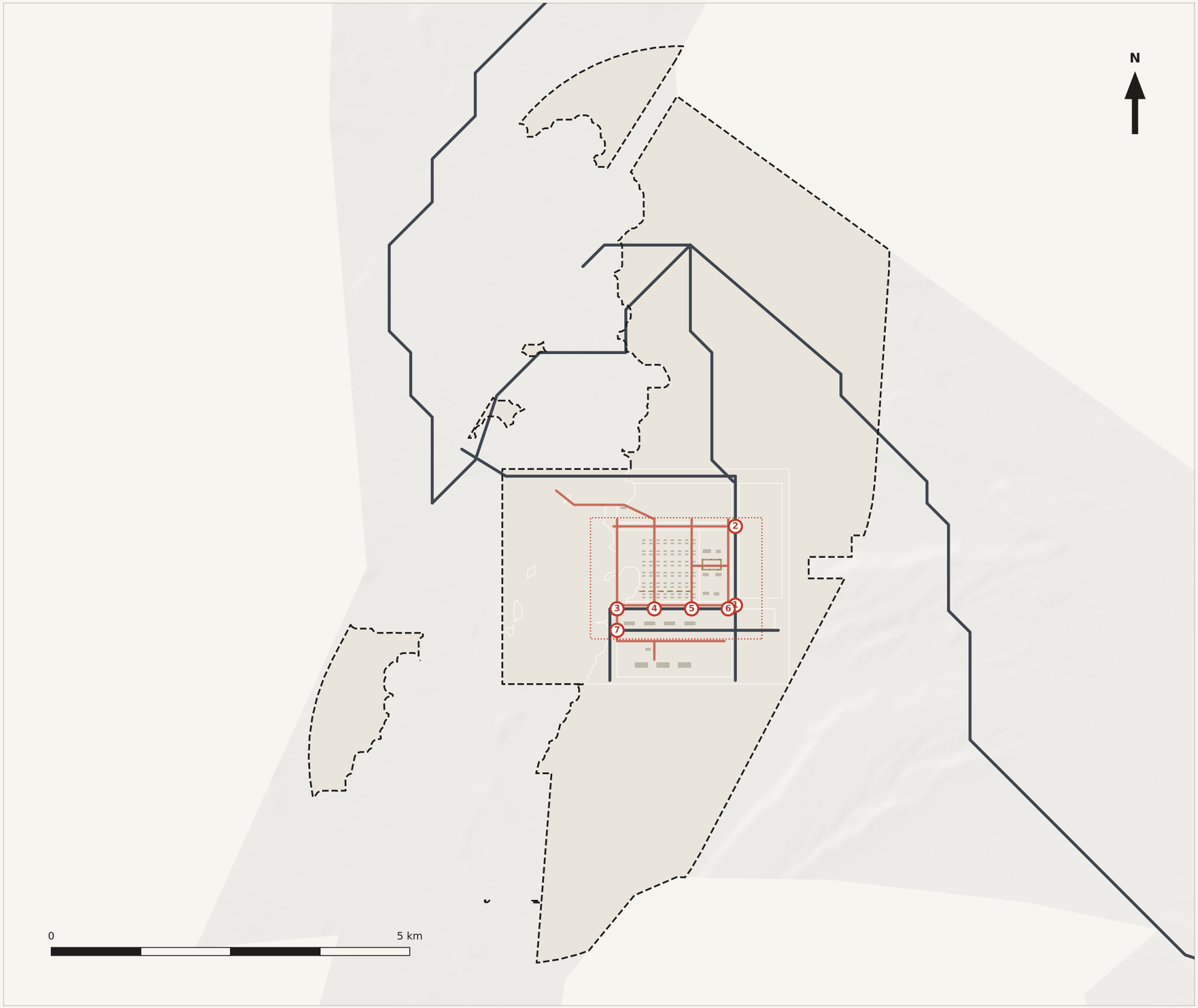
Recomputed from geometry

Coverage 0.40% · FAR 0.01 · open space 17%

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Every human-machine interface numbered



Legend

- People 13 km
- Machines + goods 158 km
- Shared 0 km
- Human-machine interface (7)

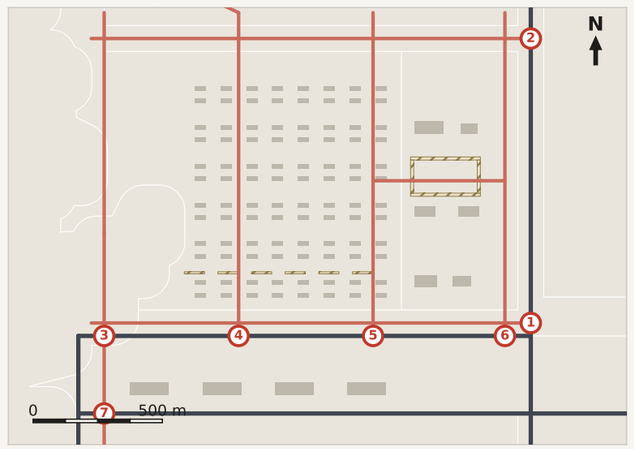
Machine traffic rule

separated

Interfaces

7 at-grade points where people and machines meet

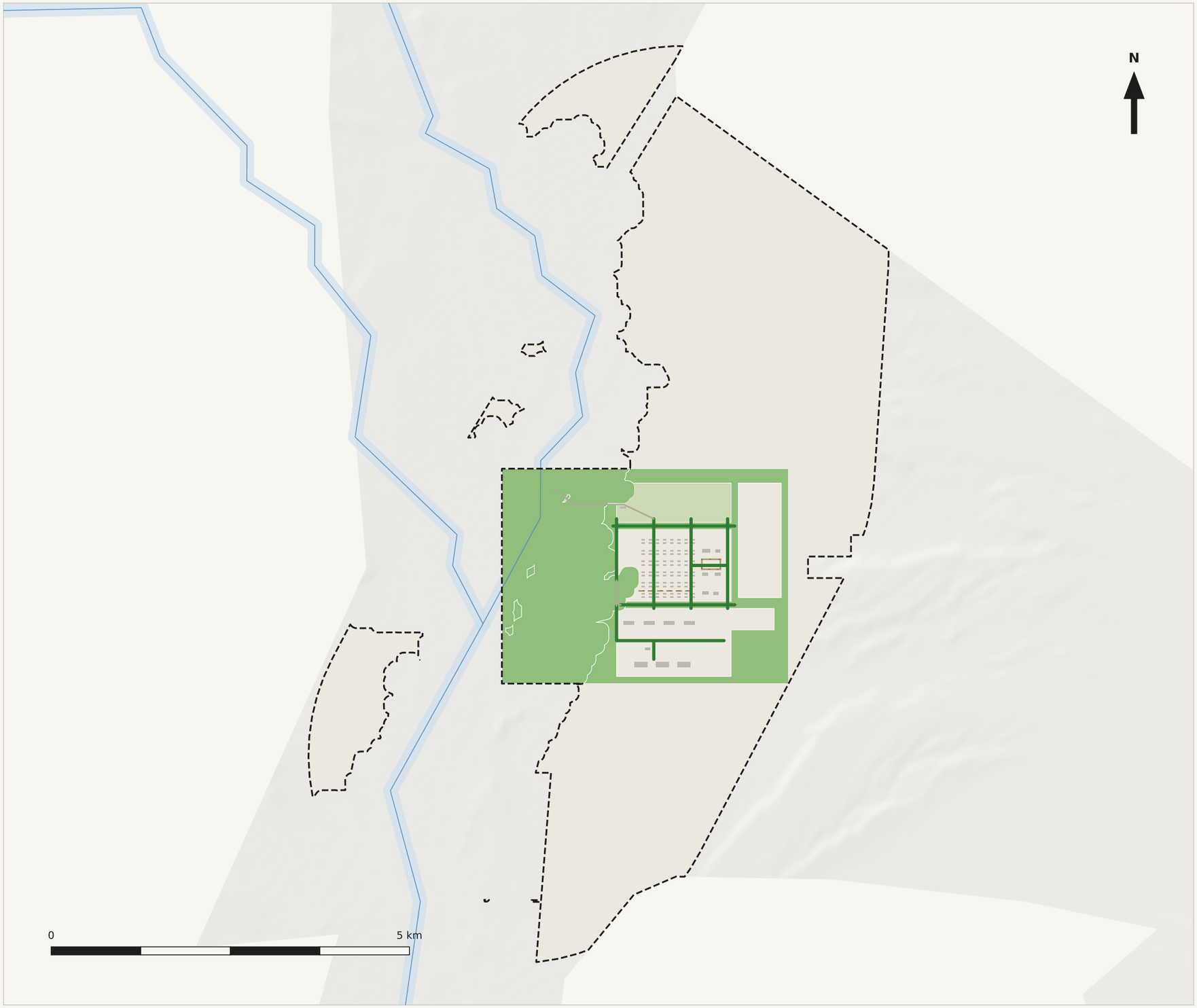
Interfaces, enlarged



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Channels are DEM-derived flow lines



Legend
- Open space 17%
- Agriculture
- Shaded walking 84%
- Unshaded walking
- Seasonal channel
- Flood corridor shown at 250 m / 100 m

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A03 · Public space in heat and wind: aerial and eye-level views

Concept views illustrate submitted space; they are not survey, capacity verification or built facts.





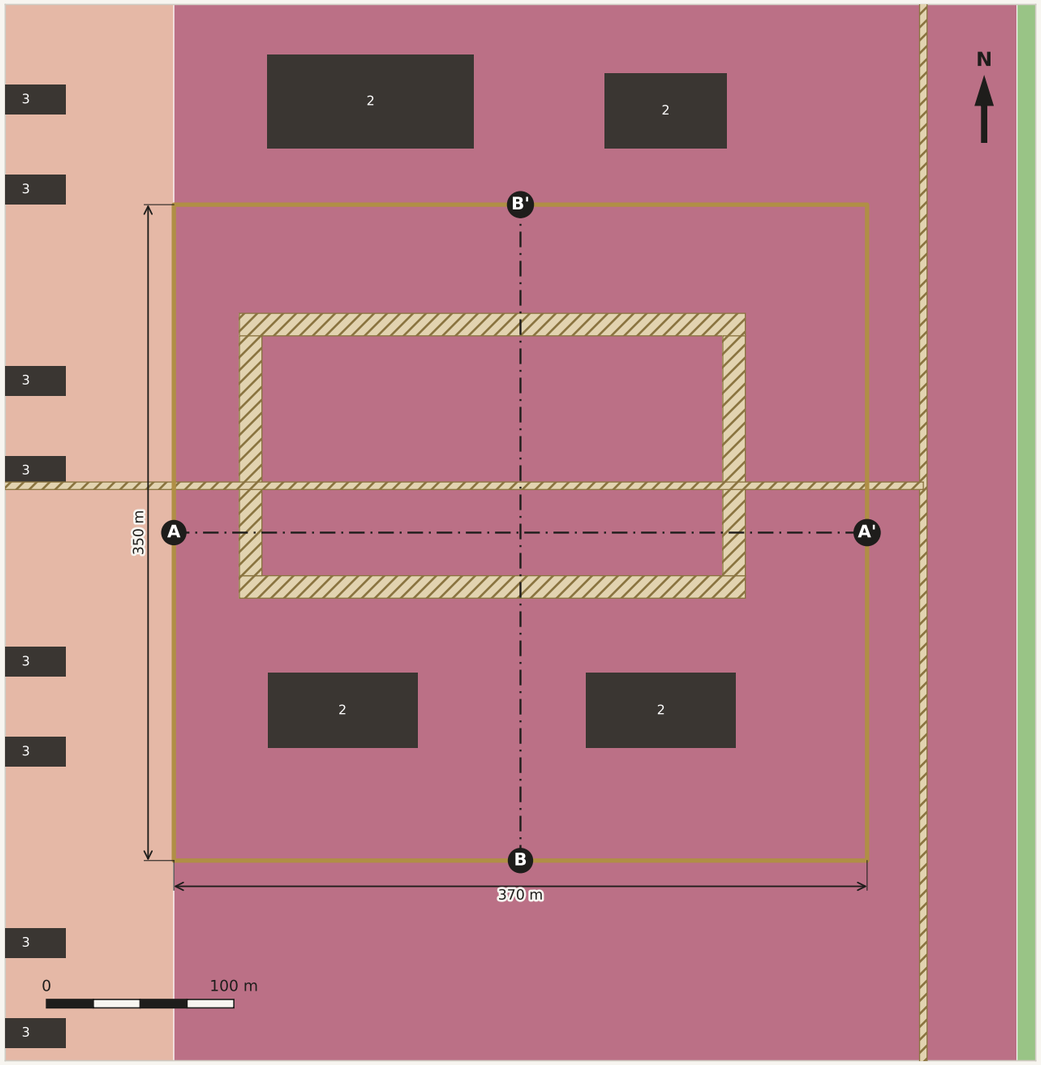
Civic core east of housing: shade, services and interaction.

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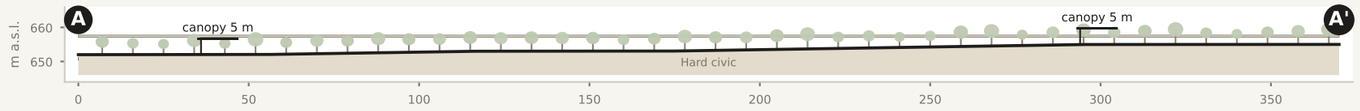
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Plan · overall dimensions, street widths, interfaces numbered, section lines

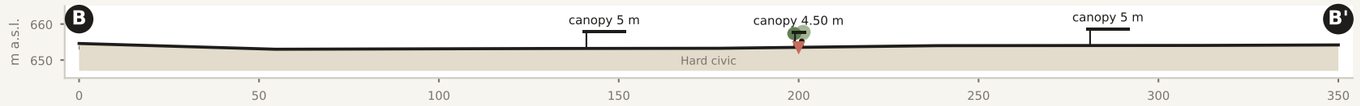


Scale 3 · dimensions and sections from the geometry

Section A-A' · 370 m · true scale, looking left of the cut



Section B-B' · 350 m · true scale, looking left of the cut



Numbers in the node · recomputed from the geometry

Node area 129,500 m² (13 ha)
Buildings 2 · footprint 6,400 m² · floor area 12,800 m²
Coverage 4.90% · FAR 0.10 · open space 0%
Floors: 2 x 2F
Floor area by use: civic 12,800 m²
Canopies (roof on posts, no floor area) 11,030 m²
Networks in the node: people 370 m
Shaded share of the walking network 100%
Human-machine interfaces 0
Trees 41 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

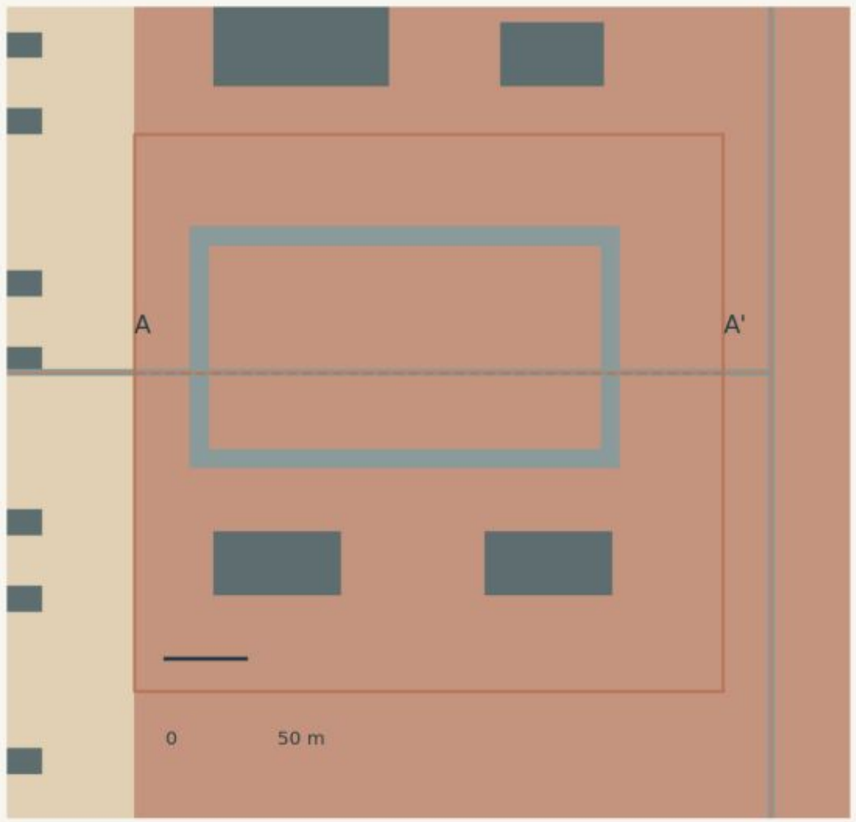
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Drawn by the entrant

A03 · Public space in heat and wind: detailed plan and decisive section

Formal plans explain spatial decisions; raw controls and extra checks stay in the technical appendix.



Occupied edges and cross-site links

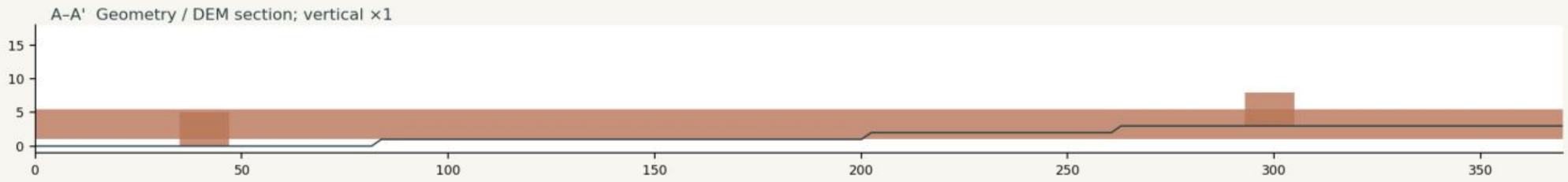
Approx. 270 × 152 m outer frame; 12 m edge roofs. Central covered walking links homes and civic facilities.

A large court is not fully shaded

Shade prioritises routes and occupied edges. Open-court activities follow heatwave operating rules.

Services sustain everyday use

Kitchen, library and training/community uses are adjacent. Drinking water, maintenance and indoor refuge remain essential.



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Drawn by the entrant

A03 · Use and operation: normal, extreme and recovery

Everyday use and abnormal operation constrain one another; unverified performance is not claimed as achieved.

Early morning

Walking, shopping and meeting

Use connected shade roofs and the public kitchen.

Heat / strong dust winds

Reduce outdoor stays

Shade does not guarantee safety; move to accessible indoor refuge.

Evening

Public activity, reading and care

Low-glare route lighting; no large screens or uplighting.

Outage / equipment failure

Keep basic access and manual service

Manual controls, emergency drinking water and staff takeover.

In every state: locate users, name the shutdown authority, provide help and verify before restarting.

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B03 · Compute and waste heat: aerial and eye-level views

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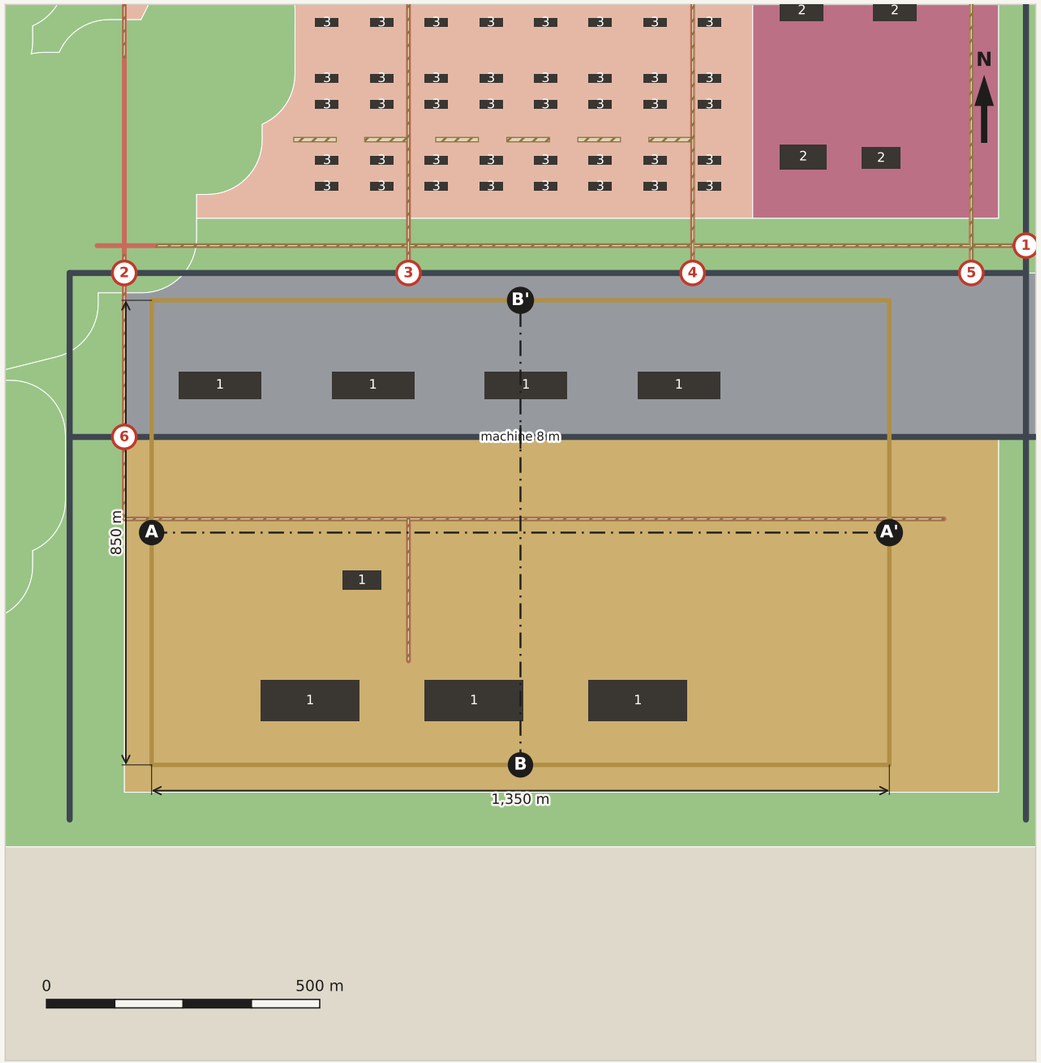
Southern production edge: compute, repair and heat receivers.

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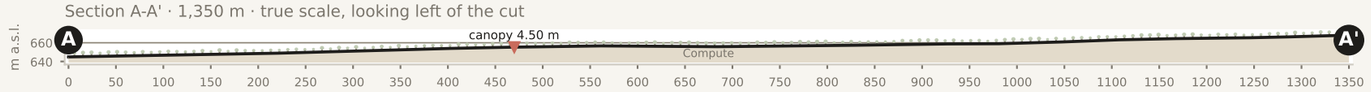
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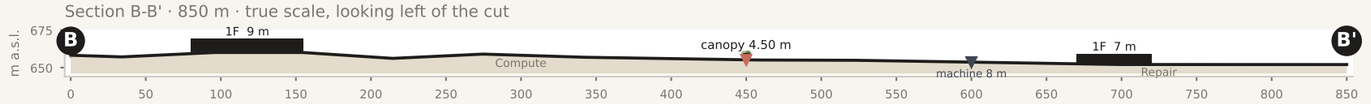


Scale 3 · dimensions and sections from the geometry

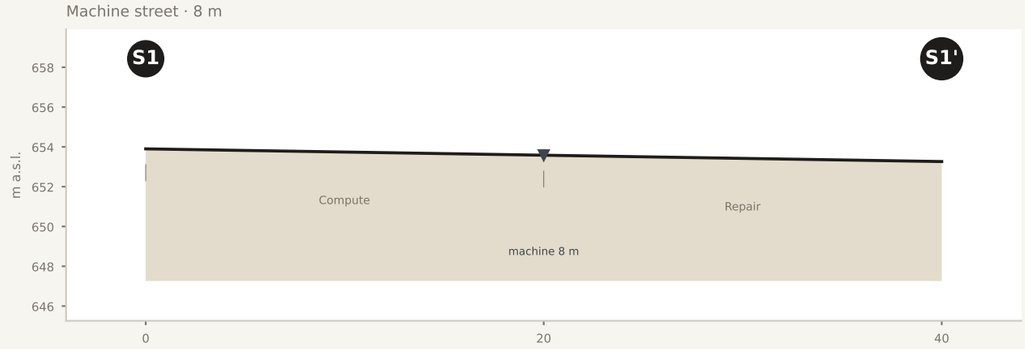
Section A-A' · 1,350 m · true scale, looking left of the cut



Section B-B' · 850 m · true scale, looking left of the cut



Machine street · 8 m



Numbers in the node · recomputed from the geometry

Node area 1,147,500 m² (115 ha)
Buildings 8 · footprint 72,950 m² · floor area 72,950 m²
Coverage 6.40% · FAR 0.06 · open space 0%
Floors: 8 x 1F
Floor area by use: compute 40,500 m², industry 32,500 m²
Canopies (roof on posts, no floor area) 6,440 m²
Networks in the node: machine 1,350 m, people 1,610 m
Shaded share of the walking network 100%
Human-machine interfaces 0
Trees 175 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

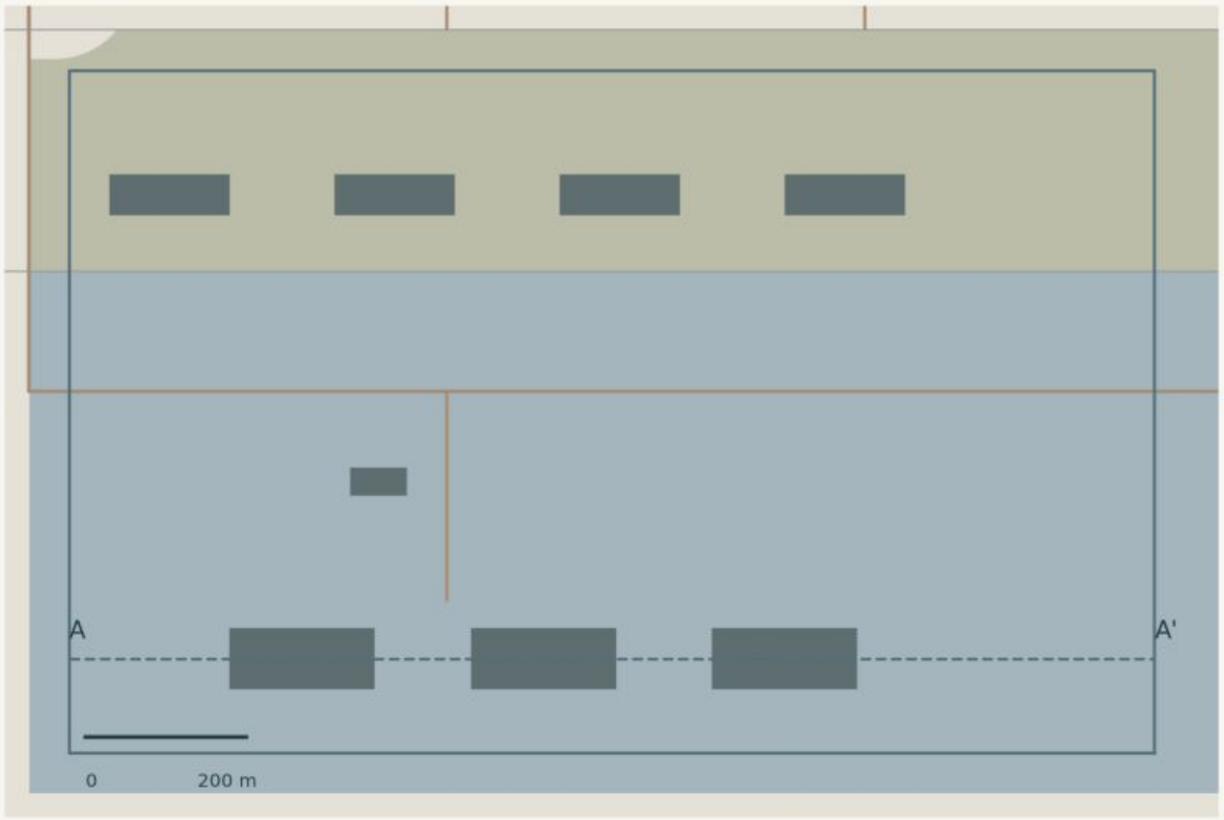
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Drawn by the entrant

B03 · Compute and waste heat: detailed plan and decisive section

Formal plans explain spatial decisions; raw controls and extra checks stay in the technical appendix.



Production layout

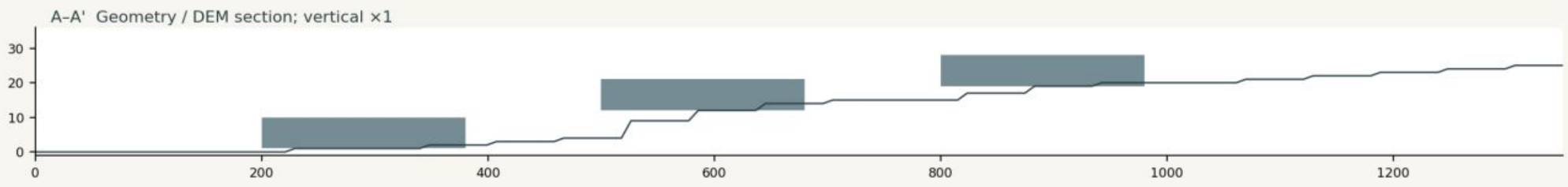
Three 180 × 75 m halls, one floor, 9 m high. Repair and processing lie to the north.

Heat needs a receiver

A 70 × 35 m drying / heat receiver is modelled. Verify network temperature, losses and operating hours.

Do not trade water for peak efficiency

Assumed targets: WUE 0.02 L/kWh, PUE 1.25. Dry rejection accepts hot-hour energy costs and curtailment.



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Drawn by the entrant

B03 · Use and operation: normal, extreme and recovery

Everyday use and abnormal operation constrain one another; unverified performance is not claimed as achieved.

Early morning

Maintenance, handover and drying preparation

Machines use the service edge; visitors follow separate paths.

Hot afternoon

Protect basic services first

Curtail flexible compute by contract, without extra cooling abstraction.

Water shortage

Stop discretionary irrigation and ornamental uses first

Protect essential resident demand and the fixed systems.

Equipment / operator exit

Transfer maintenance and assets

Keep public utilities, metering and service funds available.

In every state: locate users, name the shutdown authority, provide help and verify before restarting.

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C03 · Seasonal river park: aerial and eye-level views

Concept views illustrate submitted space; they are not survey, capacity verification or built facts.





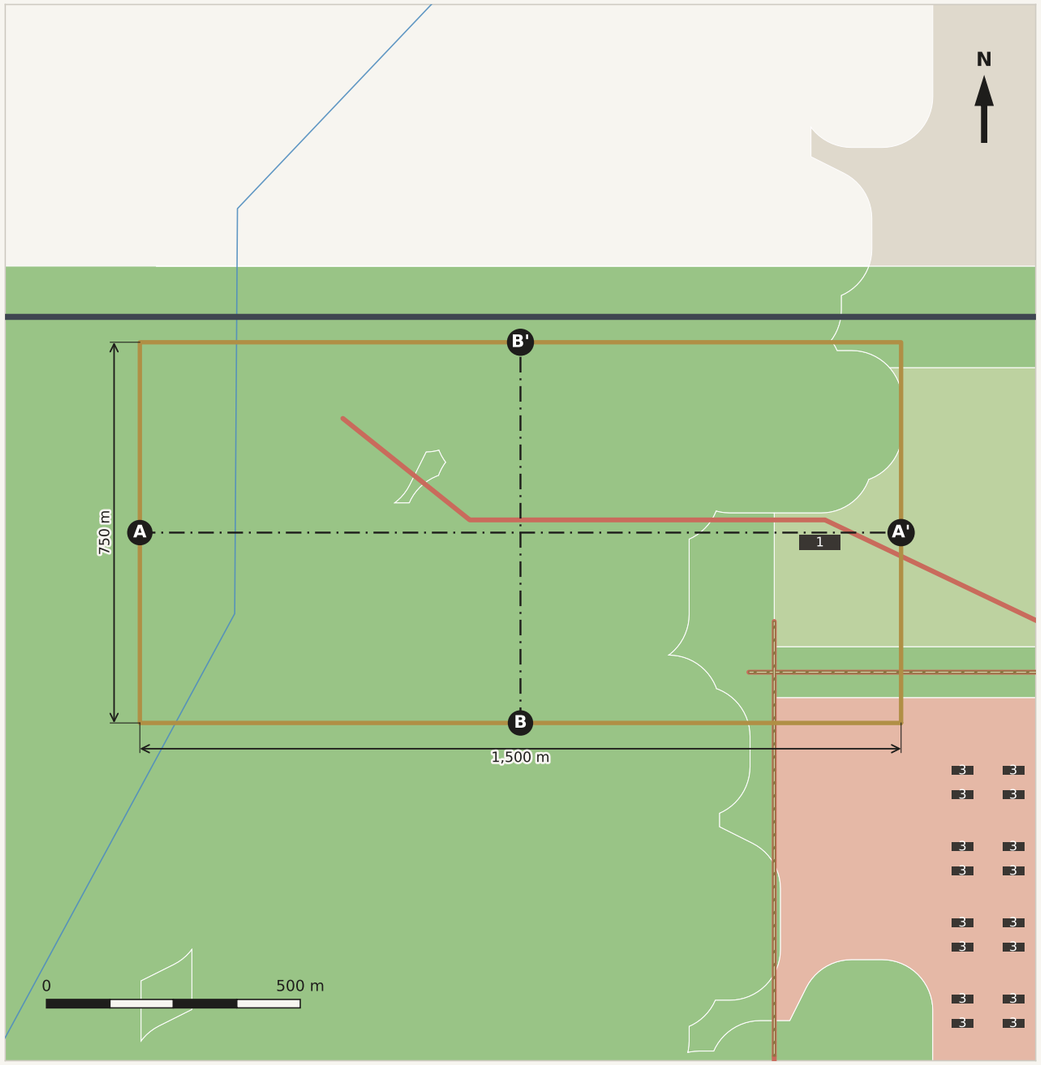
Northwest channel edge: dry everyday use and flood retreat.

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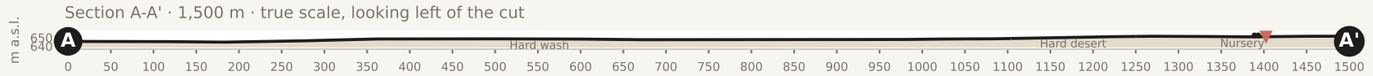
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Plan · overall dimensions, street widths, interfaces numbered, section lines



Scale 3 · dimensions and sections from the geometry

Section A-A' · 1,500 m · true scale, looking left of the cut



Section B-B' · 750 m · true scale, looking left of the cut



Numbers in the node · recomputed from the geometry

Node area 1,125,000 m² (112 ha)
Buildings 1 · footprint 2,400 m² · floor area 2,400 m²
Coverage 0.20% · FAR 0 · open space 92%
Floors: 1 x 1F
Floor area by use: civic 2,400 m²
Canopies (roof on posts, no floor area) 2,000 m²
Networks in the node: people 1,686 m
Shaded share of the walking network 30%
Human-machine interfaces 0
Trees 55 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

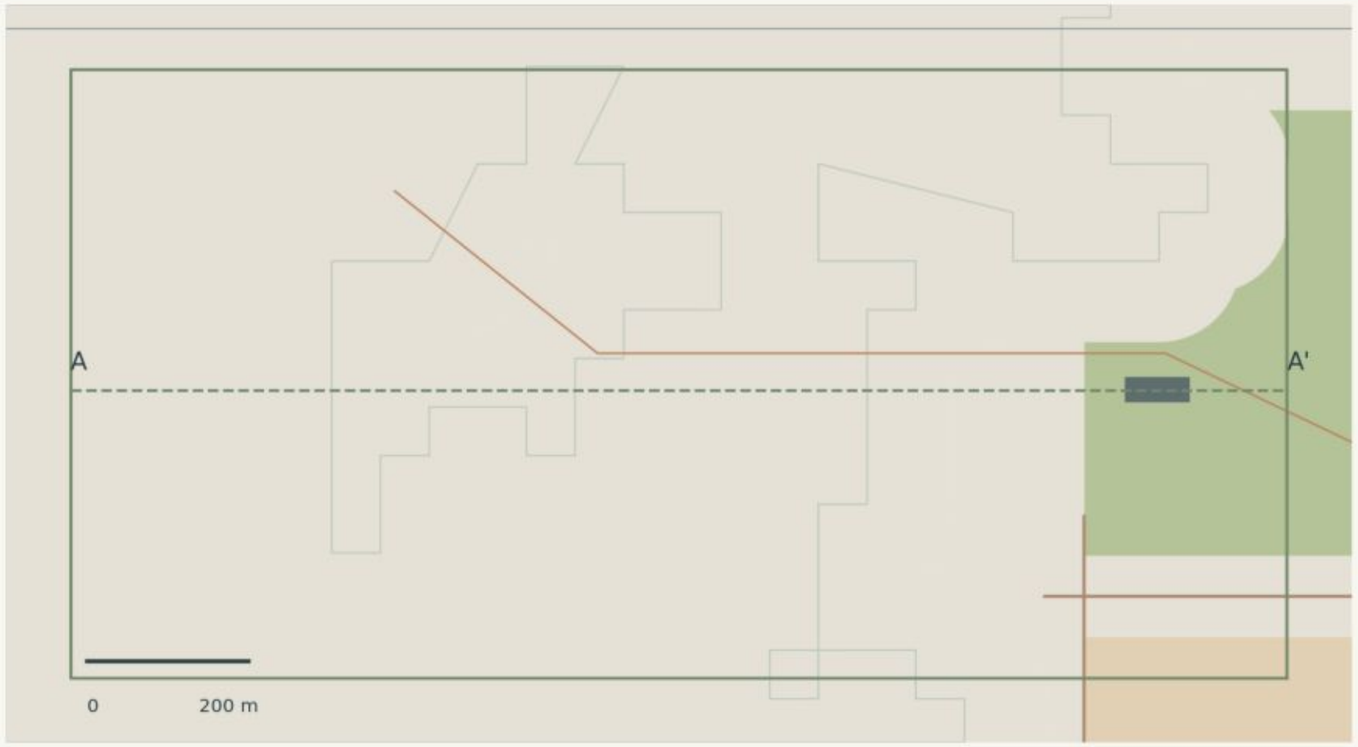
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Drawn by the entrant

C03 · Seasonal river park: detailed plan and decisive section

Formal plans explain spatial decisions; raw controls and extra checks stay in the technical appendix.



Everyday use on higher ground

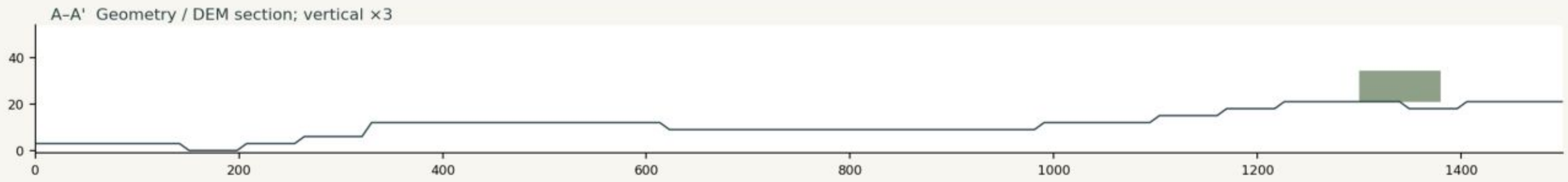
An 80 × 30 m field classroom occupies the eastern high ground. Paths observe dry channels, not a permanent lake.

A clear retreat in storms

Close low seasonal paths; preserve flood conveyance. Duty staff manage warning, closure and reopening.

A section is not flood certification

Ground comes from a 60 m DEM with vertical exaggeration. Design levels, flows and small washes are unverified.



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Drawn by the entrant

C03 · Use and operation: normal, extreme and recovery

Everyday use and abnormal operation constrain one another; unverified performance is not claimed as achieved.

Dry everyday use

Observation, walking and water education

The classroom is on high ground; channels retain their dry landscape.

Upstream warning

Close low paths and guide retreat

Duty staff control closure; do not wait for rain at the site.

During floods

Low areas closed

No design flow is supplied; the model does not certify protection.

After recession

Inspect erosion, sediment and path damage

Maintenance staff inspect before reopening; no automatic unlock.

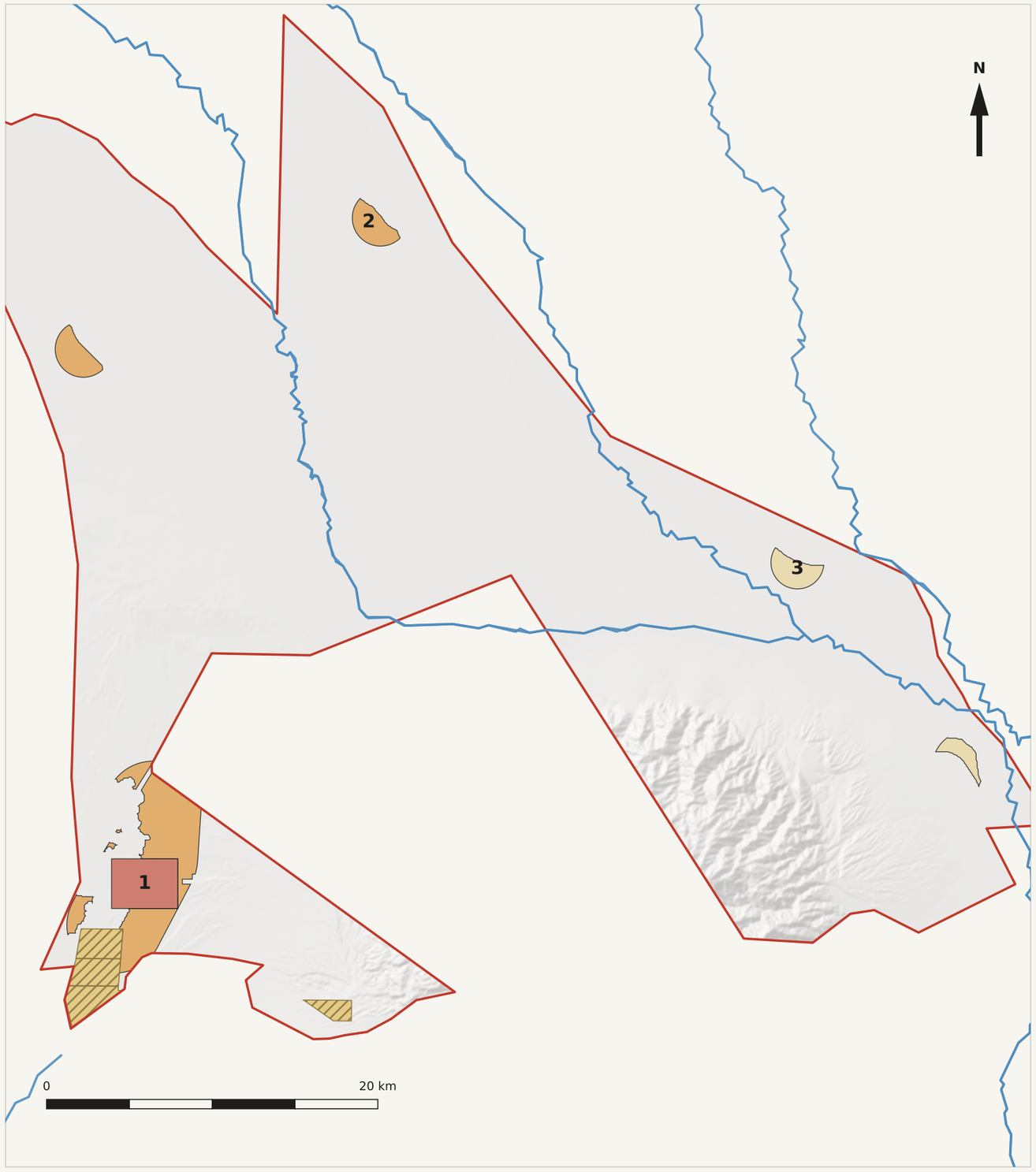
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Observable triggers, not dates



Legend
- Phase 1 12 km²
- Phase 2 38 km²
- Phase 3 7.21 km²

Phase 1 - Independent first neighbourhood and regional access works

Phase one: no trigger.

Phase 2 - Conditioned urban and productive expansion

Twelve months audited 20% reliable-water headroom; 80% occupancy; each extension has funded services, processing and safe access

Phase 3 - Eastern gateway and industry

Hydraulic and road engineering accepted; binding trade demand; lawful reliable water and power; own services and emergency access funded

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Full accounts and checks are in the separate technical appendix

Indicator	Value	Unit	Scope / phase	Evidence
Master design area	41	km2	Master area	computed
Residents	2,400	persons	Phase 1	declared
Jobs	1,200	jobs	Phase 1	declared
Homes	960	units	Phase 1	declared
Floor area ratio	0.01	ratio	Master area	computed
Open space	17	%	Master area	computed
Annual water demand	1,000,000	m3/yr	Phase 1	assumed
Compute load	12	MW	Phase 1	assumed
Hourly clean energy match	62	%	Phase 1	computed
Water reuse	20	%	Phase 1	assumed
Waste heat reuse	5	%	Phase 1	assumed

Reading these figures

The whole site is 1,656 km²; master-area figures are not whole-site totals. Phase 1 means full build-out and occupancy unless stated otherwise. Evidence tags describe the basis, not approval. Full system land accounts, assumptions and calculations remain in the appendix and metrics.json.

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Water budget: protect daily life before expanding production

Assumed gross use: 1,000,000 m³/yr. Reuse is internal transfer, not another external water source.



Agriculture + cleaning 778,263.2 m³/yr

Residents + civic services 140,160.0 m³/yr

Industry 60,000.0 m³/yr

Landscape 20,000.0 m³/yr

Compute make-up 1,576.8 m³/yr

External input

840,000 m³ fresh input less 40,000 losses: 800,000 reaches use, plus 200,000 internal reuse.

Protect the fixed agriculture allowance

600,000 m³ is an unverified pilot allowance. If actual use is higher, reduce new discretionary uses first.

What a 30% shortfall costs

Cut landscape by 20,000; new irrigation/cleaning by 178,263.2; industry by 40,000; non-essential domestic use by 61,736.8 m³.

An unresolved demand gap

Fixed living-quarter demand is absent. This budget does not cover the entire future city.

If basic demand cannot be met, secure emergency water or delay occupancy; arithmetic does not supply water.

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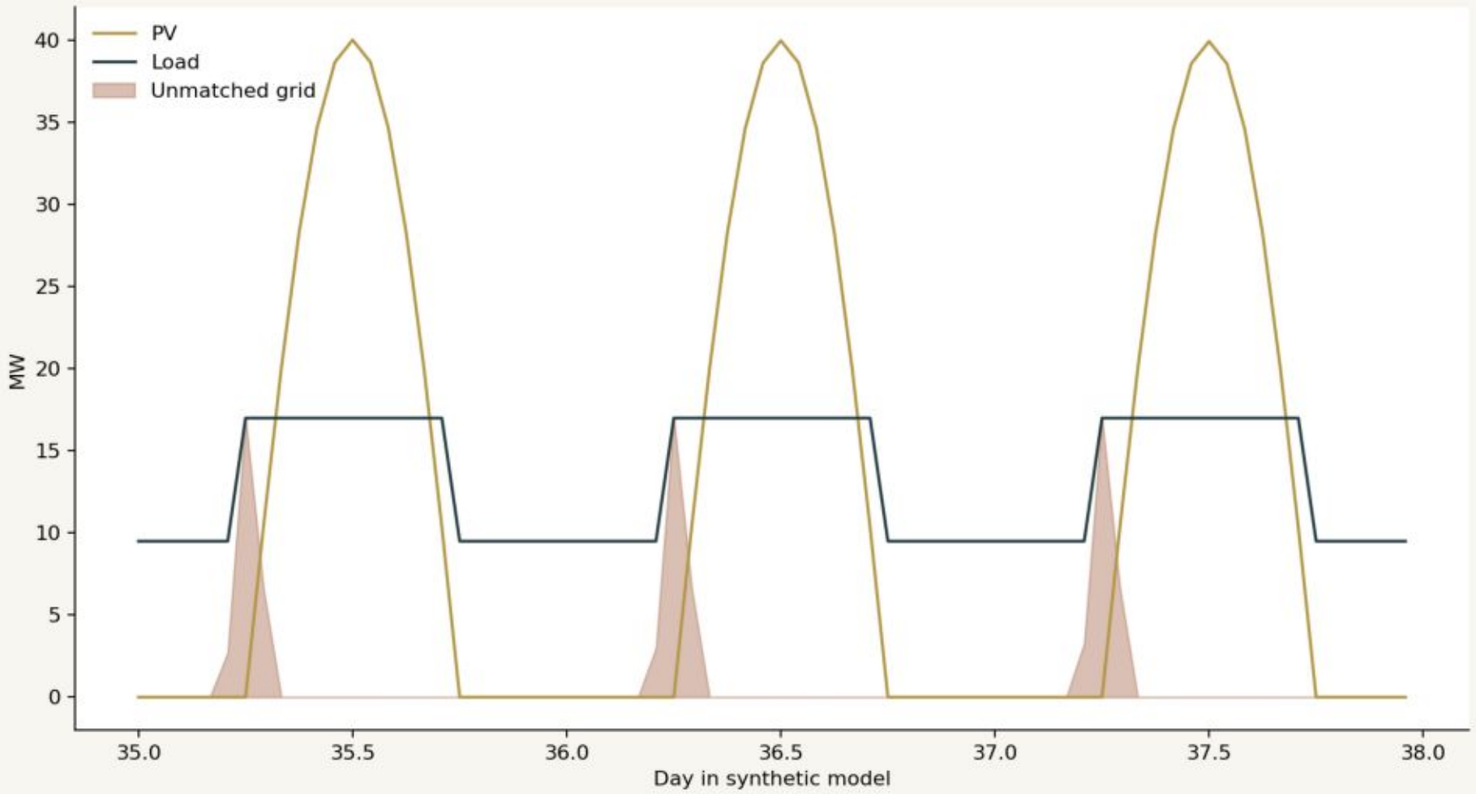
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Energy: night tests the promise more than daylight

An 8,760-hour synthetic scenario, not measured weather, a grid agreement or achieved performance.



62.432%

Share of hours fully matched by clean supply; not the annual clean-energy share.

72.993%

Separate clean-energy share. Unverified grid imports receive no clean credit.

Model inputs

Assume 40 MW PV, 480 MWh storage, 12 MW peak IT and PUE 1.25.

Storage starts at its protected reserve: 216 MWh deliverable emergency energy is not free initial clean supply.

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Drawn by the entrant

Who authorises growth and pays for downtime and maintenance?

Critical city utilities must survive any single compute operator.

Public utility + community trust

Retain land, utilities, metering and public access. Publish aggregate water, power, storage and compute accounts.

Compute operator

Contract interruptible service and maintain equipment. Provide transferable interfaces and safe exit arrangements.

Residents, workers + independent audit

Participate in service priorities and appeals without personal tracking. Independent audit checks meters, calculations and growth conditions.

AI advises dispatch; it cannot authorise expansion, alter essential water allocations or override pedestrian priority.

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